

SOLUTION

1. A* SEARCH ALGORITHM

Solution

Edge Costs

- $A \rightarrow B = 2$
- $A \rightarrow C = 4$
- $B \rightarrow C = 3$
- $B \rightarrow D = 7$
- $B \rightarrow E = 2$
- $C \rightarrow E = 3$
- $D \rightarrow E = 2$
- $D \rightarrow G = 2$

Heuristic Values

Node $h(n)$

A	7
B	6
C	4
D	3
E	2
G	0

Step 1

Current Node = A

- $g(A) = 0$
- $h(A) = 7$
- $f(A) = 7$

Expand A:

Node	g(n)	h(n)	f(n)
B	2	6	8
C	4	4	8

Open List: B(8), C(8)

Closed List: A

Step 2

Choose **B** (tie broken alphabetically).

Expand B.

Possible paths:

- $C = 2 + 3 = 5$ (Existing cost to C is 4, so ignore)
- $D = 2 + 7 = 9$
- $E = 2 + 2 = 4$

Node	g(n)	h(n)	f(n)
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C	4	4	8
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E	4	2	6
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D	9	3	12
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Open List: E(6), C(8), D(12)

Closed List: A, B

Step 3

Choose **E** (lowest f).

Expand E.

No direct edge to G.

Open List: C(8), D(12)

Closed List: A, B, E

Step 4

Choose **C**.

Expand C.

New path to E:

$$g(C) + 3 = 7$$

$$\text{Existing } g(E) = 4$$

Ignore (higher cost).

Open List: D(12)

Closed List: A, B, E, C

Step 5

Choose **D**.

Expand D.

Path to G:

$$g(D) = 9$$

$$g(G) = 9 + 2 = 11$$

Node g(n) h(n) f(n)

G	11	0	11
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Open List: G(11)

Closed List: A, B, E, C, D

Step 6

Choose **G**.

Goal reached.

Open List: Empty

Closed List: A, B, E, C, D, G

Iteration	Current Node	Open List	Closed List
1	A	B(8), C(8)	A
2	B	E(6), C(8), D(12)	A, B
3	E	C(8), D(12)	A, B, E
4	C	D(12)	A, B, E, C
5	D	G(11)	A, B, E, C, D
6	G	Empty	A, B, E, C, D, G

Optimal Path

A → B → D → G

Total Path Cost

$$2 + 7 + 2 = \boxed{11}$$

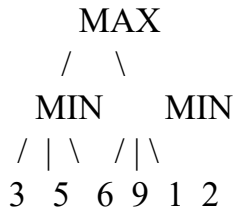
Answer:

- Optimal Path: A → B → D → G
- Total Path Cost: 11

2. MINIMAX WITH ALPHA-BETA PRUNING

Solution

Given Game Tree



Evaluate left to right.

Initial values:

- $\alpha = -\infty$
- $\beta = +\infty$

Step 1: Root (MAX)

MAX

$\alpha = -\infty$

$\beta = +\infty$

Go to the left MIN node.

Step 2: Left MIN Node

Initial:

$\alpha = -\infty$

$\beta = +\infty$

Evaluate children.

Visit 3

$\beta = \min(+\infty, 3) = 3$

Current:

$\alpha = -\infty$

$\beta = 3$

Since

$\beta(3) \leq \alpha(-\infty)?$

No.

Visit 5

$\beta = \min(3, 5) = 3$

No pruning.

Visit 6

$\beta = \min(3, 6) = 3$

Left MIN returns

Value = 3

Step 3: Back to Root (MAX)

Current value

$MAX = \max(-\infty, 3) = 3$

Update

$\alpha = 3$

$\beta = +\infty$

Go to the right MIN node.

Step 4: Right MIN Node

Initial

$\alpha = 3$

$\beta = +\infty$

Visit 9

$\beta = \min(+\infty, 9) = 9$

Check

$9 \leq 3$?

No.

Visit 1

$\beta = \min(9, 1) = 1$

Now

$\beta = 1$

$\alpha = 3$

Since

$\beta \leq \alpha$

$1 \leq 3$

YES

Alpha-Beta Cutoff occurs.

So node 2 is pruned.

Right MIN returns

Value = 1

Step 5: Back to Root

$MAX = \max(3, 1) = 3$

Final value

3

DRY RUN TABLE

Step	Current Node	α	β	Selected Value	Pruned
1	MAX	$-\infty$	$+\infty$	—	—
2	Left MIN	$-\infty$	3	3	None
3	MAX	3	$+\infty$	3	—
4	Right MIN	3	1	1	2
5	MAX	3	$+\infty$	3	—

FINAL ANSWER

Best Move for MAX

Choose Left MIN

Final Minimax Value

3

Pruned Node

2